

4 Vectors

1. Calculate the following dot and cross products of $\mathbf{u}$ and $\mathbf{v}$.

$$(a) \quad \mathbf{u} = \begin{bmatrix} 2 \\ -4 \\ \sqrt{5} \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} -2 \\ 4 \\ \sqrt{5} \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ -4 \\ \sqrt{5} \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 4 \\ \sqrt{5} \end{bmatrix} = (2)(-2) + (-4)(4) + (\sqrt{5})(\sqrt{5}) = -4 - 16 + 5 = -15$$

$$\begin{bmatrix} 2 \\ -4 \\ \sqrt{5} \end{bmatrix} \times \begin{bmatrix} -2 \\ 4 \\ \sqrt{5} \end{bmatrix} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & -4 & \sqrt{5} \\ -2 & 4 & \sqrt{5} \end{vmatrix} = \begin{bmatrix} -4\sqrt{5} - 4\sqrt{5} \\ -2\sqrt{5} - 2\sqrt{5} \\ 8 - 8 \end{bmatrix} = \begin{bmatrix} -8\sqrt{5} \\ -4\sqrt{5} \\ 0 \end{bmatrix}$$

$$(b) \quad \mathbf{u} = \begin{bmatrix} 1.5 \\ -0.5 \\ 1 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1.5 \\ -0.5 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = (1.5)(1) + (-0.5)(1) + (1)(2) = 1.5 - 0.5 + 2 = 3$$

$$\begin{bmatrix} 1.5 \\ -0.5 \\ 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1.5 & -0.5 & 1 \\ 1 & 1 & 2 \end{vmatrix} = \begin{bmatrix} -1 - 1 \\ -3 + 1 \\ 1.5 + 0.5 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \\ 2 \end{bmatrix}$$

$$(c) \quad \mathbf{u} = \begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = (2)(-1) + (-2)(1) + (4)(1) = -2 - 2 + 4 = 0$$

$$\begin{bmatrix} 2 \\ -2 \\ 4 \end{bmatrix} \times \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & -2 & 4 \\ -1 & 1 & 1 \end{vmatrix} = \begin{bmatrix} -2 - 4 \\ -2 - 4 \\ 2 - 2 \end{bmatrix} = \begin{bmatrix} -6 \\ -6 \\ 0 \end{bmatrix}$$

$$(d) \quad \mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = 0 \text{ and } \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$(e) \quad \mathbf{u} = 2\hat{\mathbf{i}} + 3\hat{\mathbf{k}} \text{ and } \mathbf{v} = \hat{\mathbf{j}} + 5\hat{\mathbf{k}}$$

$$\begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix} = (2)(0) + (0)(1) + (3)(5) = 0 + 0 + 15 = 15$$

$$\begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} \times \begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & 0 & 3 \\ 0 & 1 & 5 \end{vmatrix} = \begin{bmatrix} 0 - 3 \\ -10 + 0 \\ 2 - 0 \end{bmatrix} = \begin{bmatrix} -3 \\ -10 \\ 2 \end{bmatrix}$$

(f) $\mathbf{u} = 2\hat{\mathbf{i}}$ and $\mathbf{v} = -3\hat{\mathbf{j}}$

$$\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -3 \\ 0 \end{bmatrix} = (2)(0) + (0)(-3) + 0 = 0$$

$$\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ -3 \\ 0 \end{bmatrix} = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2 & 0 & 0 \\ 0 & -3 & 0 \end{vmatrix} = \begin{bmatrix} 0 - 0 \\ -0 + 0 \\ -6 - 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -6 \end{bmatrix}$$

2. Find the parametric and symmetric equations of each of the following lines.

(a) The line through the point $P(3, -4, -1)$ parallel to the vector $\hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$.

$$\begin{cases} x = 3 + t \\ y = -4 + t \\ z = -1 + t \end{cases} \quad \text{or} \quad \frac{x-3}{1} = \frac{y+4}{1} = \frac{z+1}{1}$$

(b) The line through $P(1, 2, -1)$ and $Q(-1, 0, 1)$.

$$\mathbf{v} = \overrightarrow{PQ} = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \\ 2 \end{bmatrix}$$

$$\begin{cases} x = 1 - 2t \\ y = 2 - 2t \\ z = -1 + 2t \end{cases} \quad \text{or} \quad \frac{x-1}{-2} = \frac{y-2}{-2} = \frac{z+1}{2}$$

(c) The line through $(2, 3, 4)$ and perpendicular to the plane $3x + 7y - 2z = 6$.

$$\mathbf{v} = \mathbf{n} = \begin{bmatrix} 3 \\ 7 \\ -2 \end{bmatrix}$$

$$\begin{cases} x = 2 + 3t \\ y = 3 + 7t \\ z = 4 - 2t \end{cases} \quad \text{or} \quad \frac{x-2}{3} = \frac{y-3}{7} = \frac{z-4}{-2}$$

3. Determine whether each the following pair of lines are parallel, intersect, or skew.

$$(a) \quad \begin{cases} x = 1 + 2t \\ y = 3 - t \\ z = 3t \end{cases} \quad \text{and} \quad \begin{cases} x = 3 + 4t \\ y = 5 - 2t \\ z = 6t \end{cases}$$

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \text{ and } \mathbf{v}_2 = \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix} \Rightarrow 2\mathbf{v}_1 = \mathbf{v}_2 \Rightarrow \text{the lines are parallel}$$

$$(b) \quad \begin{cases} x = t - 1 \\ y = 3 + t \\ z = t \end{cases} \quad \text{and} \quad \begin{cases} x = 1 - t \\ y = 3 - t \\ z = -t \end{cases}$$

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{v}_2 = \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} \Rightarrow -\mathbf{v}_1 = \mathbf{v}_2 \Rightarrow \text{the lines are parallel}$$

$$(c) \quad \begin{cases} x = 3 + 2t \\ y = -1 + 4t \\ z = 2 - t \end{cases} \quad \text{and} \quad \begin{cases} x = 1 + 4t \\ y = 1 + 2t \\ z = -3 + 4t \end{cases}$$

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 4 \\ -1 \end{bmatrix} \text{ and } \mathbf{v}_2 = \begin{bmatrix} 4 \\ 2 \\ 4 \end{bmatrix} \Rightarrow \text{the lines are not parallel, say they intersect at } (a, b, c).$$

$$\begin{cases} 3 + 2t = a = 1 + 4u \\ -1 + 4t = b = 1 + 2u \\ 2 - t = c = -3 + 4u \end{cases} \Rightarrow \begin{cases} t - 2u = -1 \\ 2t - u = 1 \\ t + 4u = 5 \end{cases} \Rightarrow t = 1, u = 1 \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \\ 1 \end{bmatrix}$$

The lines intersect at (5, 3, 1).

$$(d) \quad \begin{cases} x = 3 + 2t \\ y = -1 + 4t \\ z = 2 - t \end{cases} \quad \text{and} \quad \begin{cases} x = 3 + 2t \\ y = 2 + t \\ z = -2 + 2t \end{cases}$$

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 4 \\ -1 \end{bmatrix} \text{ and } \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \Rightarrow \text{the lines are not parallel, say they intersect at } (a, b, c).$$

$$\begin{cases} 3 + 2t = a = 3 + 2u \\ -1 + 4t = b = 2 + u \\ 2 - t = c = -2 + 2u \end{cases} \Rightarrow \begin{cases} t - u = 0 \\ 4t - u = 3 \\ t + 2u = 4 \end{cases}$$

No solution for t and u . Thus, the two lines are skew.

$$(e) \quad \begin{cases} x = 1 + 2t \\ y = -1 - t \\ z = 3t \end{cases} \quad \text{and} \quad \begin{cases} x = 2 - t \\ y = 3t \\ z = 1 + t \end{cases}$$

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \text{ and } \mathbf{v}_2 = \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix} \Rightarrow \text{the lines are not parallel, say they intersect at } (a, b, c).$$

$$\begin{cases} 1 + 2t = a = 2 - u \\ -1 - t = b = 3u \\ 3t = c = 1 + u \end{cases} \Rightarrow \begin{cases} 2t + u = 1 \\ -t - 3u = 1 \\ 3t - u = 1 \end{cases}$$

No solution for t and u . Thus, the two lines are skew.

$$(f) \quad \begin{cases} x = 2 - t \\ y = 3t \\ z = 1 + t \end{cases} \quad \text{and} \quad \begin{cases} x = 5 + 2t \\ y = 1 - t \\ z = 8 + 3t \end{cases}$$

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 3 \\ 1 \end{bmatrix} \text{ and } \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \Rightarrow \text{the lines are not parallel, say they intersect at } (a, b, c).$$

$$\begin{cases} 2 - t = a = 5 + 2u \\ 3t = b = 1 - u \\ 1 + t = c = 8 + 3u \end{cases} \Rightarrow \begin{cases} -t - 2u = 3 \\ 3t + u = 1 \\ t - 3u = 7 \end{cases} \Rightarrow t = 1, u = -2 \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix}$$

The lines intersect at $(1, 3, 2)$.

4. Find the standard equation of each of the following planes.

(a) The plane through $P(0, -2, -1)$ normal to $\mathbf{n} = 3\hat{\mathbf{i}} + 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$

$$3(x - 0) + 2(y - (-2)) - (z - (-1)) = 0$$

$$\Rightarrow 3x + 2y + 4 - z - 1 = 0 \Rightarrow 3x + 2y - z + 3 = 0$$

(b) The plane through $(1, -1, 3)$ parallel to the plane $3x + y + z = 7$

$$3(x - 1) + (y - (-1)) + (z - 3) = 0$$

$$\Rightarrow 3x - 3 + y + 1 + z - 3 = 0 \Rightarrow 3x + y + z - 5 = 0$$

(c) The plane through $(1, 1, -1)$, $(2, 0, 2)$, and $(0, -2, 1)$

$$\mathbf{n} = \left(\begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} - \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \right) \times \left(\begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} - \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} \right) = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 3 & -2 \\ 2 & 2 & 1 \end{vmatrix} = \begin{bmatrix} 7 \\ -5 \\ -4 \end{bmatrix}$$

$$7(x - 2) - 5(y - 0) - 4(z - 2) = 0$$

$$\Rightarrow 7x - 14 - 5y - 4z + 8 = 0 \Rightarrow 7x - 5y - 4z - 6 = 0$$

(d) The plane through $(2, 4, 5)$, $(1, 5, 7)$, and $(-1, 6, 8)$

$$\mathbf{n} = \left(\begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix} - \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix} \right) \times \left(\begin{bmatrix} -1 \\ 6 \\ 8 \end{bmatrix} - \begin{bmatrix} 1 \\ 5 \\ 7 \end{bmatrix} \right) = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & -1 & -2 \\ -2 & 1 & 1 \end{vmatrix} = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$$

$$(x - 1) + 3(y - 5) - (z - 7) = 0$$

$$\Rightarrow x - 1 + 3y - 15 - z + 7 = 0 \Rightarrow x + 3y - z - 9 = 0$$

(e) The plane through $P(2, 4, 5)$ perpendicular to the line $\begin{cases} x = 5 + t \\ y = 1 + 3t \\ z = 4t \end{cases}$

$$(x - 2) + 3(y - 4) + 4(z - 5) = 0$$

$$\Rightarrow x - 2 + 3y - 12 + 4z - 20 = 0 \Rightarrow x + 3y + 4z - 34 = 0$$

- (f) The plane through $A(1, -2, 1)$ perpendicular to the vector from the origin to A .

$$\mathbf{n} = \overrightarrow{OA} = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

$$(x - 1) - 2(y - (-2)) + (z - 1) = 0$$

$$\Rightarrow x - 1 - 2y - 4 + z - 1 = 0 \Rightarrow x - 2y + z - 6 = 0$$

5. Find the distance of each of the following.

- (a) The point $(0, 0, 12)$ and the line $\begin{cases} x = 4t \\ y = -2t \\ z = 2t \end{cases}$

$$Q(0, 0, 0) \text{ is on the line} \Rightarrow \overrightarrow{PQ} \times \mathbf{v} = \begin{bmatrix} 0 \\ 0 \\ -12 \end{bmatrix} \times \begin{bmatrix} 4 \\ -2 \\ 2 \end{bmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & -12 \\ 4 & -2 & 2 \end{vmatrix} = \begin{bmatrix} -24 \\ -48 \\ 0 \end{bmatrix}$$

$$d = \frac{\|\overrightarrow{PQ} \times \mathbf{v}\|}{\|\mathbf{v}\|} = \frac{24\sqrt{(-1)^2 + (-2)^2}}{2\sqrt{2^2 + (-1)^2 + 1^2}} = 12\sqrt{\frac{5}{6}} \approx 10.9545 \text{ units.}$$

- (b) The point $(2, 1, 3)$ and the line $\begin{cases} x = 2 + 2t \\ y = 1 + 6t \\ z = 3 \end{cases}$

$$Q(2, 1, 3) \text{ is on the line} \Rightarrow d = 0$$

- (c) The point $(0, 1, 3)$ and the plane $2x + y + 2z = 4$

$$d = \frac{|2(0) + 1(1) + 2(3) - 4|}{\sqrt{2^2 + 1^2 + 2^2}} = \frac{3}{\sqrt{9}} = 1 \text{ unit}$$



- (d) The line $\begin{cases} x = 2 + t \\ y = 1 + t \\ z = -0.5 - 0.5t \end{cases}$ and the plane $x + 2y + 6z = 10$

$$\mathbf{n} \cdot \mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ -0.5 \end{bmatrix} = (1)(1) + (2)(1) + (6)(-0.5) = 1 + 2 - 3 = 0$$

Since the line is 'parallel' to the plane, the distance can be measured by calculating the distance between the plane and the point $(2, 1, -0.5)$ on the line.

$$d = \frac{|1(2) + 2(1) + 6(-0.5) - 10|}{\sqrt{1^2 + 2^2 + 6^2}} = \frac{|2 + 2 - 3 - 10|}{\sqrt{41}} = \frac{9}{\sqrt{41}} \approx 1.4056 \text{ units}$$

6. Determine the name of each of the following quadric surfaces.

- (a) $9x^2 - y^2 - z^2 = 9 \Rightarrow x^2 - \frac{y^2}{9} - \frac{z^2}{9} = 1$ Hyperboloid of two sheets
- (b) $9x^2 + y^2 + z^2 = 9 \Rightarrow x^2 + \frac{y^2}{9} + \frac{z^2}{9} = 1$ Ellipsoid
- (c) $z = x^2 + 4y^2$ Elliptic paraboloid
- (d) $x^2 + 4y^2 - z^2 = 1$ Hyperboloid of one sheet
- (e) $z = x^2 - 4y^2$ Hyperbolic paraboloid
- (f) $z^2 = x^2 + 4y^2$ Cone

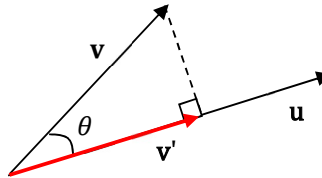


7.

Given two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$, let $\mathbf{v}'$ be the projection vector of $\mathbf{v}$ on $\mathbf{u}$. Show that

$$\mathbf{v}' = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|^2} \right) \mathbf{u}$$

Let θ be the angle between $\mathbf{u}$ and $\mathbf{v}$.



Note that

$$\|\mathbf{v}'\| = \begin{cases} \|\mathbf{v}\| \cos \theta & \text{for } 0 \leq \theta < \frac{\pi}{2} \\ -\|\mathbf{v}\| \cos \theta & \text{for } \frac{\pi}{2} < \theta \leq \pi \end{cases}$$

and since $\mathbf{v}'$ is parallel to $\mathbf{u}$,

$$\hat{\mathbf{v}}' = \begin{cases} \hat{\mathbf{u}} & \text{for } 0 \leq \theta < \frac{\pi}{2} \\ -\hat{\mathbf{u}} & \text{for } \frac{\pi}{2} < \theta \leq \pi \end{cases}$$

So,

$$\frac{\mathbf{v}'}{\|\mathbf{v}'\|} = \hat{\mathbf{u}} \Rightarrow \mathbf{v}' = \frac{\|\mathbf{v}'\|}{\|\mathbf{u}\|} \mathbf{u} = \left(\frac{\|\mathbf{v}\| \cos \theta}{\|\mathbf{u}\|} \right) \mathbf{u} = \left(\frac{\|\mathbf{u}\| \cdot \|\mathbf{v}\| \cos \theta}{\|\mathbf{u}\|^2} \right) \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|^2} \right) \mathbf{u}$$